Importance of Camber Angle

1. At the time of turning, camber angle ensures the ease of steering.
2. To reduce the tyre wear.
3. To reduce the load acting on the king pin and the wheel bearing due to weight of the vehicle.

8.9.2.3 King Pin Inclination

When the vehicle is viewed from front, the angle between the inclination of the king pin from the vertical axis is called the King pin Inclination. The inclination is normally kept 7° to 8°.

The reason for king pin inclination

1. It gives good road holding on turnings.
2. It reduces steering effort.
3. It reduces the load acting on the wheel bearing.

8.9.2.4 Toe-in and Toe-out

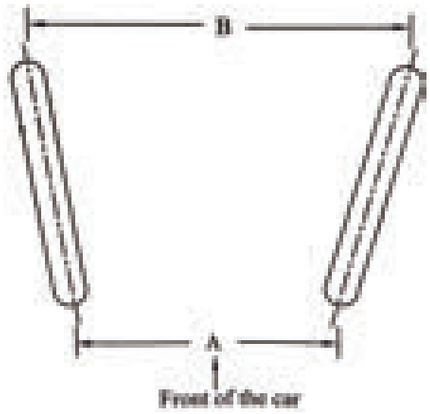
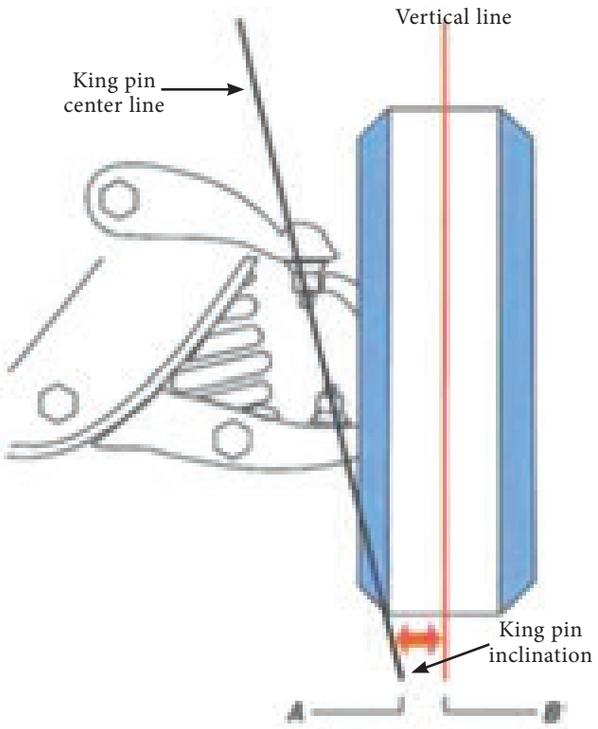


Figure 8.9.2.4 Toe-in

Toe-in

When viewed from the top, the distance between the front wheels at the front is less than the distance at the back is called Toe-in. Normally toe-in is up to 3 mm.



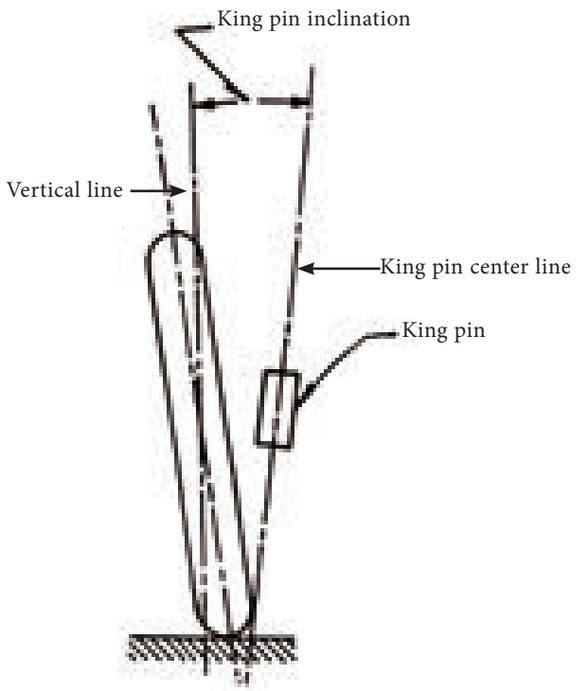


Figure 8.9.2.3 (a) & (b) King pin inclination

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